

Conversational Paradigm Systems: Integrating Rule-Based Dialogue, RAG, and MCP for NASA Telemetry

*Natural Language Interaction (ILN) 2026/2027 - Master in Artificial Intelligence
Deliverable 3 (Option 1: Integration of Agency and Detailed Evaluation)*

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Abstract

This project investigates how rule-based dialogue, Retrieval-Augmented Generation (RAG), intent classification, and external tool use can be integrated within a single conversational architecture. The system combines a rule-based chatbot for student well-being with a NASA-focused RAG agent using Sentence-BERT embeddings, ChromaDB, and 17 NASA technical reports, while an intent classifier routes queries to the appropriate component. For Deliverable 3, the RAG agent was extended through the Model Context Protocol (MCP), enabling access to structured reasoning tools, current NASA data, and astronomical observations from the STScI MAST archive. The integrated system was evaluated using a 20-question NASA benchmark and a five-question benchmark focused on live MCP tool use. Results show that RAG and MCP improve grounding and citation behaviour and are particularly useful for queries requiring current information, although they introduce additional response latency and do not outperform the baseline on every metric. These findings illustrate both the benefits and practical trade-offs of combining multiple conversational paradigms in one system.

Keywords: Conversational Systems, Retrieval-Augmented Generation, Model Context Protocol, Tool Use, Intent Classification, Aerospace.

1 Introduction, Motivation, and Objectives

Natural Language Interaction (NLI) has developed from early rule-based and pattern-matching systems to modern conversational systems based on Large Language Models (LLMs) [Vaswani et al., 2017]. These approaches offer different advantages. Rule-based systems provide predictable behaviour and allow developers to control how particular inputs are handled, but their ability to respond to previously unseen language is limited. LLM-based systems are considerably more flexible, but they can generate incorrect or unsupported information and their internal

knowledge may become outdated [Lewis et al., 2020; Es et al., 2023]. These limitations are particularly important when a conversational system is expected to answer technical questions or provide information that may change over time. This project investigates how different conversational approaches can be combined within a single system. Rather than relying on one technique for every type of interaction, different components are used according to the type of query received. The project was developed through the three deliverables specified for the Natural Language Interaction course, with each stage adding additional capabilities to the system.

Deliverable 1 - Rule-Based Well-Being Agent. The first component, The Optimist, is a rule-based conversational agent designed primarily for student well-being and general conversation. It extends regular-expression matching with linguistic preprocessing, including lemmatization and dependency parsing. This provides a predictable conversational component while allowing some variation in how users express similar intentions.

Deliverable 2 - NASA RAG Expert Agent. The second component is a Retrieval-Augmented Generation (RAG) agent specialising in NASA space missions and aerospace topics. It uses Sentence-BERT embeddings and a ChromaDB vector database containing information extracted from official NASA technical reports. Relevant document sections are retrieved for each query and supplied to the LLM as context, with the aim of producing answers grounded in the source material and providing document and page references. To combine these two different conversational approaches, an intent classification router was developed using Sentence-BERT embeddings and Logistic Regression. Incoming queries are classified as either general/well-being conversation or aerospace-related questions and directed to the appropriate agent. Queries for which the classifier has insufficient confidence trigger a clarification request rather than being automatically assigned to an agent.

Deliverable 3 - MCP Tool Integration. For the third deliverable, the RAG agent was extended using the Model Context Protocol (MCP) [Anthropic, 2024]. Three MCP servers were integrated to provide additional capabilities: Sequential Thinking for structured multi-step processing,

NASA public APIs for current mission and space data, and the STSci MAST archive for astronomical observations. This allows the expert agent to supplement information retrieved from the static document collection with information obtained from external services when required. The main objective of the final system is therefore to evaluate whether rule-based dialogue, RAG, intent classification, and external tool use can be combined effectively within a single conversational architecture. Particular attention is given to answer quality, grounding and citation behaviour, access to current information, routing accuracy, and response latency. The evaluation also examines cases where the additional complexity of RAG and MCP does not improve performance, allowing the advantages and limitations of the different approaches to be compared.

2 Detailed Description of Implemented Agents

The complete system architecture synthesizes four modular subsystems: The Intent Router, the Rule-Based Engine (D1), the Expert RAG Vector Store (D2), and the Agentic MCP Tri-Server Orchestrator (D3-O1).

2.1 Deliverable 1: 'The Optimist' Rule-Based Persona

Deliverable 1 (implemented in D1rule.py) establishes an empathetic, cheerful conversational companion designed to assist university students working through academic stress, exam anxiety, and emotional burnout. While traditional NLTK chat agents rely on brittle surface string matching, our engine incorporates modern linguistic normalization to achieve robust lexical generalization under 0.5 ms execution latency:

- 1. spaCy Linguistic Normalization Pipeline:** Incoming text is tokenized and lemmatized using spaCy's `en_core_web_sm` model. Rather than authoring hundreds of combinatorial regex patterns for every verb inflection ('studying', 'studied') and plural noun ('exams', 'deadlines'), the text is reduced to lowercase canonical lemmas ('study', 'exam', 'deadline') and filtered for punctuation. Custom grammatical reflections map pronouns across conversational perspectives ('i am' -> 'you are', 'my' -> 'your').
- 2. Prioritized Regex Rules with Negation Handling:** Regular expressions are evaluated in strict priority order. Negation patterns ('not happy', 'cannot cope', 'never good') are prioritized over affirmative emotion rules, preventing catastrophic misclassifications where a student stating 'I am not happy with my grades' would otherwise match an affirmative pattern. Affective keywords ('overwhelmed', 'failing', 'exhausted', 'thesis') trigger therapeutic, evidence-based coping recommendations.
- 3. Dynamic Context Buffering & Topic Extraction Fallback:** Maintains conversation state across turns, recording user attributes (name, degree program, imminent deadlines). When an utterance triggers no

explicit regex, an intelligent fallback parses the dependency tree to extract prominent NOUN and PROP heads, prompting the user with contextual follow-ups regarding their topic.

2.2 Semantic Intent Routing & Polysemy Handling

To integrate the rule-based agent (D1) and the NASA expert agent (D2), `D1D2classifierrouter.py` implements an intent classifier that determines which agent should handle each user query. The router uses the same 384-dimensional Sentence-BERT model (all-MiniLM-L6-v2) [Reimers and Gurevych, 2019] used by the RAG system, avoiding the need to load an additional embedding model.

Feature Representation and Classification. Each incoming utterance is converted into a normalized 384-dimensional sentence embedding $e(x)$. A Logistic Regression classifier is then used to classify the query as either CHIT_CHAT (D1) or NASA_EXPERT (D2). The classifier was trained on a manually labelled dataset of 66 utterances containing student well-being and general conversational queries together with technical aerospace questions:

$$P(y = c | x) = \frac{\exp(w_c^T e(x) + b_c)}{\sum_j \exp(w_j^T e(x) + b_j)} \quad (1)$$

Confidence Threshold and Ambiguous Queries. Some queries may contain vocabulary that could reasonably relate to either agent. For example, the word stress could refer to emotional stress experienced by a student or structural stress on a spacecraft. Rather than always selecting the class with the highest probability, the router uses a confidence threshold of $\tau = 0.60$. If the highest predicted probability is at least 0.60, the query is sent to the corresponding agent. Otherwise, the system asks the user to clarify whether the query concerns general/well-being conversation or an aerospace topic:

$$p_{\max}(x) = \max_c P(y = c | x),$$

$$\text{Action}(x) = \begin{cases} \arg \max_c P(y = c | x), & p_{\max}(x) \geq 0.60 \\ \text{DISAMBIGUATE}, & \text{otherwise} \end{cases} \quad (2)$$

Table 1 shows examples of the router's behaviour. Queries such as "I feel an immense amount of stress about exams" are directed to D1, while "What aerodynamic stress does a rocket experience at Max-Q?" is directed to D2. This demonstrates how the sentence embeddings allow the classifier to use the context of the complete utterance rather than relying only on individual keywords. The reported confidence values range from 66.8% for the Apollo 11 alarm query to 87.3% for the loneliness query.

Test Context & Utterance	Routed Intent	Confidence
'I feel an immense amount of stress about exams'	CHIT_CHAT (D1)	84.4%
'What aerodynamic stress does rocket experience at Max-Q?'	NASA_EXPERT (D2)	69.8%

Test Context & Utterance	Routed Intent	Confidence
'I need to work on my study habits and motivation'	CHIT_CHAT (D1)	74.2%
'How does the ChemCam laser work on Mars targets?'	NASA_EXPERT (D2)	82.4%
'Can you help me feel less lonely today?'	CHIT_CHAT (D1)	87.3%
'Explain cryogenic sunshield of James Webb telescope'	NASA_EXPERT (D2)	85.5%
'My laptop crashed and I am so upset'	CHIT_CHAT (D1)	69.5%
'What caused Apollo 11 to trigger 1201 alarms?'	NASA_EXPERT (D2)	66.8%

Table 1: Router Polysemous Boundary Disambiguation Performance

2.3 Deliverable 2: Expert NASA Space Missions RAG Agent

Deliverable 2 (implemented in D2RAG.py) establishes an authorizing knowledge retrieval and generation engine specializing in NASA space exploration missions:

- **Corpus Ingestion & Domain Coverage:** An automated pipeline downloads and processes 17 official technical reports from the NASA Technical Reports Server (NTRS), covering flagship missions: JWST science instruments and cryogenic performance; Curiosity MSL and Perseverance EDL, ChemCam, and SHERLOC; Ingenuity Mars helicopter aerodynamics; Artemis SLS Core Stage RS-25 cryogenic propulsion; DART kinetic impactor telemetry; Hubble Space Telescope SM3A servicing; and Apollo 11 Saturn V flight data.
- **Dynamic Embedding-Based Semantic Chunking:** Rather than segmenting text at arbitrary, static character windows, D2RAG.py implements true semantic chunking (semantic_chunk_text). Extracted text is split into sentences and encoded in a single batched pass with all-MiniLM-L6-v2. The pipeline computes cosine distances d_i between adjacent sentences. A dynamic breakpoint threshold θ_{85} is determined by the 85th percentile of distances across the document. Sentences are accumulated into cohesive units, segmenting when a topic shift occurs ($d_i \geq \theta_{85}$, with $\min_chunk_chars = 150$) or when approaching the embedding ceiling ($\max_chunk_chars = 800$). This produced 1,861 semantically cohesive chunks with metadata tracking (filename, page_number, mission_domain).
- **Vector Database:** Persistent ChromaDB collection using cosine similarity over normalized embeddings.
- **Top-K Retrieval & Grounded Generation:** Top-K = 8 excerpts are retrieved in ~40 ms and assembled into an authoritative context block. The prompt enforces strict grounding: all claims must cite official documents in the exact format [Document_Name.pdf, Page X]. Citations are deterministically cross-referenced against chunk metadata to prevent hallucinated filenames.

2.4 Deliverable 3 (Option 1): Tri-Server Model Context Protocol (MCP) Agency

Deliverable 3 transforms the static RAG pipeline into an autonomous, tool-using agent adhering to the Model Context Protocol (MCP) specification [Anthropic, 2024]. The orchestrator (mcp_client_manager.py) bridges Ollama tool-calling representations with three live, decoupled MCP tool servers:

- **1. Sequential Thinking Server (sequential_thinking_server.py):** Implements dynamic multi-step cognitive reasoning for complex problem decomposition. Allows the agent to formulate hypotheses, evaluate evidence across chunks, revise assumptions, and branch reasoning trees. Critically, to prevent small 7B models from entering repetitive loops, effective planned thoughts are clamped to 2, enforcing immediate grounded synthesis upon plan completion.
- **2. NASA Public APIs Server (nasa_mcp_server.py):** Connects the agent to live authenticated NASA endpoints (api.nasa.gov): nasa_near_earth_objects queries NeoWs for live asteroid close-approach tracking with automatic 7-day date window clamping and /neo/browse fallback; nasa_mars_rover_manifest retrieves active sol counts and camera manifests; nasa_mars_rover_photos routes Curiosity queries to api.nasa.gov and Perseverance queries to NASA JPL raw feeds; nasa_space_weather_donki fetches Coronal Mass Ejection (CME) and solar flare alerts.
- **3. STScI MAST Astrophysics Server (mast_mcp_server.py):** Interfaces with the Mikulski Archive for Space Telescopes: mast_resolve_target resolves celestial identifiers (e.g., 'TRAPPIST-1') to equatorial coordinates (RA, Dec) via SIMBAD/NED; mast_jwst_observations queries the CAOM catalog for active JWST instrument observations, filters, and proposals.

3 Experimental Validation & Problematic Cases

The integrated system was subjected to rigorous end-to-end conversational validation across varied dialogue intents, demonstrating seamless inter-agent routing, tool calling, and deterministic fallbacks.

3.1 End-to-End Dialogue Interaction Traces

Table 2 reports real runtime execution traces recorded from the unified system, demonstrating the operational flow from user utterance to router classification, component activation, latency, and output synthesis.

Turn & Intent	User Query & Component	Latency & Behavior
T1: Well-Being	'Overwhelmed with thesis deadlines' -> D1 (The Optimist)	0.4 ms: Empathetic validation

Turn & Intent	User Query & Component	Latency & Behavior
T2: Static RAG	'4 core instruments in JWST ISIM?' -> D2 (NASA Expert)	13.8 s: Cites NIRCam, NIRSpec, MIRI, NIRISS [JWST_Payload.pdf, p.2].
T3: Live Telemetry	'Hazardous asteroids this week?' -> D3 (NeoWs Tool)	14.3 s: Queries live NeoWs; retrieves actual close-approach bodies.
T4: Deep Space	'JWST observed TRAPPIST-1?' -> D3 (MAST Tool)	12.2 s: Resolves coordinates; queries CAOM Proposal 1225.
T5: Polysemous	'Help me analyze stress?' -> Router Disambig.	1.2 ms: Clarifies emotional well-being (D1) vs structural stress (D2).

Table 2: End-to-End Conversational Traces Across Architectural Paradigms

4 Detailed Evaluation & Empirical Benchmark

For Deliverable 3 (Option 1), we conducted a quantitative evaluation comparing the ungrounded parametric baseline (Without-RAG) with the full RAG and MCP-enabled system (With-RAG + MCP). The evaluation focused on factual content, numerical telemetry, citation grounding and response latency.

4.1 Benchmark Methodology & Evaluation Metrics

The evaluation harness (`new_nasa_rag_evaluator.py`) was executed across the 20-question authoritative aerospace dataset (`nasa_eval_dataset.json`) covering JWST, DART, MSL Curiosity, Perseverance, Ingenuity, Artemis SLS, and Apollo 11. The harness computes both deterministic factual metrics and multi-dimensional LLM-as-a-Judge qualitative evaluations [Zheng et al., 2023]:

- **Fact Recall (%)**: Lexical and semantic presence of ground-truth physical constants, subsystem names, and acronyms:

$$\text{Fact Recall} = \frac{|F_{\text{candidate}} \cap F_{\text{reference}}|}{|F_{\text{reference}}|} \quad (4)$$

- **Telemetry Metric Coverage (%)**: Exact numerical verification of physical quantities, temperatures (K), velocities (km/s), masses, and orbital periods against official NASA mission reports:

$$\text{Telemetry Coverage} = \frac{|T_{\text{candidate}} \cap T_{\text{reference}}|}{|T_{\text{reference}}|} \quad (5)$$

- **Inline Citation Density & Alignment**: Frequency of verified citations adhering to the [Document.pdf, Page X] format and their exact cross-document alignment with ground-truth source PDFs.
- **Latency Decomposition**: Granular profiling separating ChromaDB retrieval, MCP tool execution, and autoregressive LLM generation.

- **LLM-as-a-Judge Rubric**: Independent qualitative scoring on a 1–5 Likert scale across Factual Accuracy, Completeness, Groundedness, and Scientific Precision using `mistral-small:24b` evaluated at temperature 0.0 with JSON structured outputs.

4.2 Full 20-Question Benchmark Results

Table 3 summarizes the empirical results across the full 20-question aerospace benchmark suite, reflecting the latest evaluation run recorded in `data/nasa_eval_results.md`:

Metric Dimension	With-RAG + MCP	Without-RAG	Delta (Δ)
Fact Recall %	14.2%	19.9%	-5.8%
Telemetry Coverage %	5.0%	6.2%	-1.2%
Average Citations / Answer	2.20	0.00	+2.20
Average Latency (s)	14.56 s	8.38 s	+6.18 s
Total MCP Tool Calls Executed	48 calls	0 calls	+48
Sequential Cognitive Thoughts	44 thoughts	0 thoughts	+44
Active MCP Tools Utilized	5 tools	None	+5
LLM Judge: Factual Accuracy	3.05 / 5.0	3.15 / 5.0	-0.10
LLM Judge: Groundedness	3.60 / 5.0	3.25 / 5.0	+0.35
LLM Judge Overall Score	3.19 / 5.0	3.40 / 5.0	-0.21

Table 3: Full 20-Question Aerospace Benchmark Results (`data/nasa_eval_results.md`)

The 20-question benchmark produced mixed results. The Without-RAG baseline achieved higher Fact Recall (19.9% vs 14.2%), Telemetry Coverage (6.2% vs 5.0%), factual accuracy (3.15 vs 3.05) and overall LLM-judge score (3.40 vs 3.19). In contrast, RAG+MCP produced an average of 2.20 citations per answer and achieved higher groundedness (3.60 vs 3.25).

The higher Fact Recall of the baseline may partly result from its tendency to produce broader answers containing more general information, increasing the likelihood of matching expected terms. The RAG system was more restricted by the retrieved context. These results therefore do not show an overall improvement from RAG+MCP on static questions; instead, its clearest advantage was improved citation and grounding. This also came at a cost, with average latency increasing from 8.38 s to 14.56 s.

4.3 Live 5-Question MCP Specialized Benchmark Suite

To evaluate real-time telemetry and active tool agency beyond static PDFs, a dedicated 5-question suite (`data/mcp_eval_results.md`) was executed with live NASA APIs and STScI MAST endpoints:

Query & Subsystem	Fact Recall (RAG/Base)	Telemetry (RAG/Base)	Latency / Cites
MCP_Q01: NeoWs Asteroids	71% / 71%	50% / 25%	25.2 s (1 cite)

Query & Subsystem	Fact Recall (RAG/Base)	Telemetry (RAG/Base)	Latency / Cites
MCP_Q02: Mars Perseverance	88% / 38%	75% / 75%	20.3 s (1 cite)
MCP_Q03: MAST TRAPPIST-1	88% / 88%	75% / 50%	18.4 s (1 cite)
MCP_Q04: DONKI Solar CME	75% / 62%	100% / 33%	25.1 s (3 cites)
MCP_Q05: DART Deflection	56% / 78%	25% / 25%	29.3 s (3 cites)
Mean Performance	75.4% / 67.3%	65.0% / 41.7%	23.7 s (1.8 avg)

Table 4: Live Agentic MCP Benchmark Results (data/mcp_eval_results.md)

The benefits of MCP were clearer on these queries. Mean Fact Recall increased from 67.3% to 75.4%, while Telemetry Coverage increased from 41.7% to 65.0%. The strongest improvements occurred for Perseverance and DONKI queries, where the system could obtain information directly from external services. However, MCP did not improve every query: for the DART question, Fact Recall was lower than the baseline (56% vs 78%).

These results suggest that MCP is most useful when the answer depends on current or externally retrieved information rather than for every technical question. This benefit also increases response time, with the five MCP queries averaging approximately 23.7 s.

4.4 Deep Scientific Analysis of Evaluation Findings

- 1. Baseline Hallucinations:** On MCP_Q01, the baseline generated unsupported asteroid designations, while the MCP system queried NeoWs and returned records from the live API, including 138971 (2001 CB21) and 2009 DC12. Similarly, for MCP_Q03, the baseline incorrectly stated that no public JWST observations of TRAPPIST-1 were available, while the MAST query returned relevant observations. Deliverable3_Report
- 2. Access to Current Information:** On MCP_Q02, the baseline returned an incorrect Perseverance mission-duration value. Using nasa_mars_rover_manifest, the MCP system retrieved current mission information and achieved 88% Fact Recall compared with 38% for the baseline. Deliverable3_Report
- 3. Sequential Tool Use:** On the DART query, the system used sequential processing to separate the problem into detection, kinetic-impact physics and the momentum enhancement factor:

$$\Delta P = \beta m v_{\text{imp}} = m v_{\text{imp}} + p_{\text{ejecta}} \quad (6)$$

Although this demonstrates multi-step tool use, the current evaluation does not isolate the contribution of Sequential Thinking itself. A separate comparison with and without this component would be required to determine whether it improves answer quality.

- 4. Grounding–Latency Trade-off:** On live queries, MCP improved mean Telemetry Coverage from 41.7% to

65.0%, but increased response time. This suggests that external tools are most appropriate when current information is required, while static RAG may be preferable when the document collection already contains sufficient information.

4.5 Detailed Case Studies and Qualitative Error Analysis

Case Study 1 – Planetary Defence (MCP_Q01): The baseline generated unsupported asteroid identifiers. The MCP agent instead queried nasa_near_earth_objects, retrieving six asteroid records and identifying 138971 as potentially hazardous, with a reported maximum diameter of 1,164.23 m and relative velocity of 36,821.98 km/h.

Case Study 2 – Space Weather (MCP_Q04): The system combined DONKI data with information from the RAG collection, achieving 75% Fact Recall and 100% Telemetry Coverage. This demonstrates how live API information can complement technical information already contained in the document collection.

Case Study 3 – DART (NASA_Q05): The system used two sequential steps to examine the momentum enhancement factor and ejecta recoil, retrieving supporting information from the DART documents. The response included the reported −33 minute orbital-period change and required 13.31 s to generate. However, the lower Fact Recall obtained for DART in the MCP benchmark also shows that additional tool use does not automatically improve every response.

5. Conclusion, Main Takeaways, Challenges and Limitations

This project designed, implemented and evaluated a multi-paradigm conversational system combining the three course deliverables. The final system integrates rule-based dialogue (D1), a NASA-focused RAG agent (D2), intent classification, and MCP-based external tool access (D3).

- 1. Integration of Conversational Approaches:** The project demonstrates that a rule-based agent and a RAG-based expert agent can be combined through a Sentence-BERT intent classifier. This allows general and well-being conversations to be handled separately from technical aerospace queries while still providing a single interface.
- 2. RAG and MCP Tool Integration:** MCP extends the RAG agent by providing access to external NASA and MAST services. This was particularly useful for questions requiring current information that was not available in the static document collection.
- 3. Evaluation Results:** The evaluation showed mixed results. On the static 20-question benchmark, RAG+MCP improved citation use and groundedness but did not outperform the baseline on all factual metrics. The benefits were clearer on the specialised live benchmark, where Fact Recall increased from 67.3% to 75.4% and Telemetry

Coverage from 41.7% to 65.0%. These improvements came with increased response latency.

5.1 Future Directions

Future work could investigate hybrid dense-sparse retrieval, combining BM25 and Sentence-BERT using Reciprocal Rank Fusion, particularly for technical identifiers that may not be represented well by semantic embeddings. A larger and more varied evaluation dataset would provide stronger evidence of system performance. The system could also be extended to determine whether MCP tools are actually required before invoking them, reducing unnecessary tool calls and response latency.

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